Differences in Disability Insurance Allowance Rates

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Abstract

Allowance rates for disability insurance applications vary by race and ethnicity, but it is unclear to what extent these differences are artifacts of other differing socio-economic and health characteristics, or selection issues in SSA's race and ethnicity data. This paper uses the 2015 American Community Survey linked to 2015-2019 SSA administrative data to investigate DI application allowance rates among non-Hispanic White, non-Hispanic Black, non-Hispanic Asian, non-Hispanic American Indian/Alaska Native, and Hispanic applicants aged 25-65. The analysis uses regression, propensity score matching, and inverse probability weighting to estimate differences in allowance rates among applicants who are similar on observable characteristics. Relative to raw comparisons, differences by race and ethnicity in multivariate analyses are substantially smaller in magnitude and are generally not statistically significant.

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Introduction

In the aggregate, non-Hispanic (NH) White applicants to Social Security Disability Insurance (DI) are approved at higher rates than NH Black and Hispanic applicants. This pattern is persistent over time and holds for both male and female applicants (GAO 1992, 2004; Godtland et al. 2007; Wang and Yang 2023). However, it is unclear whether these differences are artifacts of other between-group differences. Little research has sought to discern the magnitude of racial and ethnic inequality in disability allowance rates, net of potential confounders, among the working-age population.

A challenge for researchers studying racial and ethnic differences in DI allowance rates is the limitations of SSA data on race (Martin 2016). SSA race and ethnicity data standards have changed over time, and reporting is voluntary, which means that these data are inconsistent and include many missing values. Since the mid-1980s, most new SSN applications are from enumeration at birth, which does not collect race and ethnicity data. Because of these limitations, there is considerable selection into what race and ethnicity data are available, if any, which may bias statistics generated from these data.

Given the limitations of the SSA administrative data on race, researchers studying SSA programs at large more often use race and ethnicity data from surveys such as the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) or the Survey of Income and Program Participation (SIPP), which have been linked to SSA administrative records. These surveys also have richer data on the social and economic circumstances of respondents than what is available in the administrative records alone. But since these surveys have modest sample sizes, it is difficult to examine relatively small groups like SSDI applicants without pooling decades of data.

This paper addresses these challenges by using data from the American Community Survey (ACS), which was recently linked to SSA administrative records. Like the CPS ASEC and SIPP, the ACS uses a race and ethnicity classification scheme that is more detailed than what is available in SSA administrative records alone. However, the ACS sample, with 3.5 million addresses, is much larger than other surveys, such as the approximately 100,000 addresses in the CPS ASEC and the 14-65,000 addresses in the SIPP. Moreover, the ACS has a higher response rate, mitigating concerns about nonresponse bias. By linking the ACS to SSA administrative records, it is possible to use ACS variables, including race and ethnicity, to examine the demographic, social, and economic characteristics of disability applicants.

Our analysis combines logistic regression, which other research on the topic has used (GAO 1992, 2004), with propensity score matching (PSM) and inverse probability weighting (IPW). The latter two methods allow us to construct comparison groups of NH White applicants that are similar to NH Black, NH Asian, NH American Indian/Alaska Native (AIAN), and Hispanic applicants, separately by sex. Applicants who differ by race or ethnicity may also differ by other characteristics, which may be related to DI allowance rates. These methods account or adjust for these differences. While PSM and IPW are sometimes used to draw causal inferences from observational data, our application of these methods is a controlled descriptive analysis (Li and Li 2023).

While there are large differences by race and ethnicity in the raw application data, the analysis shows that differences in disability allowance rates between matched, reweighted, or regression-adjusted race and ethnicity groups are much smaller and usually not statistically significant. Where we find differences that are statistically significant, they are sensitive to estimation method. Taken together, these results suggest that differences in allowance rates by race and ethnicity in the raw, unadjusted data are mainly due to compositional differences between groups.

The paper is organized as follows. First, we report rates of self-reported disability, DI application, and DI allowance by race and ethnicity and briefly describe applicant characteristics. Second, we review the small body of research on racial and ethnic inequality in disability allowance races. Third, we discuss the limitations of SSA administrative data and how the matched ACS-administrative data address these limitations. Fourth, we describe the analysis dataset and our estimation methods. Fifth, we present our results. We conclude by discussing caveats of the analysis and possible directions for additional research.

Aggregate differences in application and allowance rates by race and ethnicity

Table 1 reports self-reported disability, DI application, and allowance rates by race and ethnicity for our 2015 ACS sample matched to administrative data, which we will discuss in detail in later sections.[1] We note that self-reported disability in the ACS is a broader definition of disability than work-limiting disability that is required for DI eligibility.

There are two main empirical patterns in Table 1. First, there is considerable variation between race-ethnicity groups for all outcomes. NH Black and NH AIAN individuals have the highest rates of self-reported disability and DI application, but the allowance rates for these groups are the lowest (among applicants). NH Asian and Hispanic individuals have the lowest rates of self-reported disability and DI application, while NH White individuals are the median group for self-reported disability and DI application. Second, the variation in application rates is not proportional to the incidence of self-reported disability. For example, NH Black individuals report disability about 30 percent more often than NH White individuals but apply about 70 percent more often, while NH AIAN individuals report disability about 70 percent more often than NH White individuals but apply about 40 percent more often.

Table 2 reports applicant characteristics by race and ethnicity, separately for male and female applicants. The table shows that applicant characteristics vary across groups. NH Black and Hispanic applicants of both sexes and NH AIAN males tend to be somewhat younger than the overall means. There is large variation in recent earnings and marital status. Each application has a "body system code" corresponding to the applicant's affliction (e.g., malignancies) or affected body system (musculoskeletal, respiratory, etc.). Here, too, there is variation across groups. For example, applications from NH Asian women and NH White men and women are more often filed for malignancies, which have a high allowance rate.

The DI allowance rate: the importance of considering application propensity

The disability application allowance rate[2] is comprised of a numerator -- the count of allowed applications -- and a denominator -- the count of all applications. When studying racial and ethnic differences in disability allowance rates, it is important to consider both the numerator and denominator, along with the factors that influence them.

SSA evaluates several factors when adjudicating a given application, including the applicant's impairment, severity of the impairment, work history, education, and training. The weight given to each of these factors varies by applicant age. For younger applicants, evaluators consider the severity of the impairment and whether the applicant can do *any* other work. For older applicants, evaluators also

[1] The results presented in this paper were approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008). This paper is released to inform interested parties of ongoing research and to encourage discussion of work in progress. The views expressed on statistical issues are those of the authors and not necessarily those of the Census Bureau or SSA.

[2] Strictly speaking, this is a proportion, not a rate, but it is usually referred to as a rate.

consider the applicants' education, training, and whether the applicant is likely to be successfully retrained for different work.

For an application to be allowed or denied, it must be initiated. Therefore, it is also important to consider the DI allowance rate's denominator – the count of all applications initiated – and the factors that shape the propensity to apply. The incidence of disability varies by race, ethnicity, geography, and education level (Ross and Bateman 2018), so these aspects of socio-economic status are likely to be correlated with the propensity to apply and therefore the denominator of the DI allowance rate. Spouses and families often provide assistance to disabled individuals, so if these familial supports are absent, the propensity to apply may be greater. Social Security benefit amounts are progressive: DI payments replace a large share of income for those with low earnings, while the maximum benefit amount will replace a small fraction of a high earner's income. Plausibly, this progressivity increases the propensity to apply for DI income receipt for low-income individuals on the margin of disability allowance (Autor and Duggan 2006). Arguably, variation in the propensity to apply may be more consequential than the disability determination process for between-group differences in allowance rates.

Existing research on racial and ethnic differences in disability allowance rates

There is relatively little extant empirical research on racial and ethnic differences in disability insurance allowances. In response to a request from Congress, GAO (1992) examines 1988 administrative data on DI applications. With logistic regression analyses of initial application decisions, the authors conclude that the lower allowance rate among Black applicants is largely due to other observable differences, specifically that Black applicants applied with less severe conditions and at younger ages. However, at the hearing level of the appeals process, Black applicants were less likely than White applicants to be allowed disability benefits, and these differences could not be explained by conditions, age, or other factors. GAO (2004)[3] examines data from SSA quality assurance reviews of administrative law judge (hearing-level appeal) decisions, finding no evidence of differences in decisions among applicants not represented by attorneys. However, among those applicants who are represented by attorneys, Black applicants were significantly more likely to be denied than White applicants. Both GAO (1992) and GAO (2004) use only administrative data and therefore have little information on the applicants' broader social and economic context, which may shape the propensity to apply. Moreover, both studies use SSA race and ethnicity data, which have some notable limitations that we discuss in the next section.

More recently, Wang and Yang (2023) examine differences in application and allowance rates among Black and White respondents in the Health and Retirement Study (HRS) linked to SSA administrative data. Applying machine learning models and ordinary least squares regression, the authors find no evidence of differences in allowance rates between Black and White individuals. However, there are caveats of Wang and Yang's analysis. The authors only examine Black/White differences due to limited information on race in HRS, which only has Black, White, and a residual "other" category. The HRS only studies respondents aged 50 and older, so the findings do not generalize to the DI worker population overall. Finally, HRS has a relatively low match rate to administrative records: 69 percent in Wang and Yang's analysis vs. 85-95 percent in Census Bureau surveys. The authors show that there are key differences between matched individuals and the sample overall.

Other research on the disability application and determination process has included race as a control variable while focusing on other aspects of the process. Benitez-Silva et al. (1999) and Kreider and Riphahn (2000) find that race is not significantly correlated with disability awards at the initial

[3] See also Godtland et al. (2007) for related research.

application and appeal levels. Hu et al.'s (2001) sequential model of the disability determination process shows that younger (under 35) Black applicants are more likely to be denied, but the authors do not investigate racial or ethnic inequality in other ways.

SSA race and ethnicity data versus household survey data

SSA's race and ethnicity data have two important limitations (Martin 2016, Fisher and Keane forthcoming), which together pose a challenge to studying race and ethnicity differences in disability allowance rates using only administrative data. First, there is selection into whether race and ethnicity data are available. Historically, most race and ethnicity data have been collected when individuals apply for or make changes to their Social Security card, but since 1987, the application process has most often occurred through enumeration at birth, which does not collect race and ethnicity information. Race and ethnicity data collection for migrants also has gaps. Lawful, permanent residents are enumerated at entry, but the Department of State does not provide race or ethnicity information to SSA. This selection by age and nativity biases statistics generated from these data. A second challenge is that race and ethnicity coding schemes have changed over time, and earlier coding schemes are often inadequate for present-day race and ethnicity research. Before 1980, SSA used three racial categories -- White, Black, and other – and a large share of today's DI application population is coded using this scheme. In 1980s, a new five-category, mutually exclusive scheme was implemented: Non-Hispanic Black, Non-Hispanic White, Asian/Pacific Islander, American Indian/Alaska Native, and Hispanic or Latino. However, there is selection into which classification scheme is used for an individual's race and ethnicity data.

Given the limitations of SSA administrative data on race and ethnicity, researchers studying inequality in SSA programs often use household surveys such as the CPS ASEC or SIPP matched to SSA administrative records. The CPS ASEC and SIPP use race and ethnicity coding schemes that align with OMB standards. However, it is necessary to pool many years of the surveys to investigate topics like disability allowance rates, and even with pooling, the available number of observations remain limited. Moreover, the CPS ASEC and SIPP do not survey group quarters housing, which misses an important section of the population receiving income from Social Security programs.

This paper addresses these limitations by using data from the American Community Survey, which was recently matched to SSA administrative records. The matched ACS has five significant advantages over other data sources. First, the ACS race and ethnicity data follow OMB standards that separate race and ethnicity.[4] Second, the ACS does not have the selection issues that SSA race and ethnicity data have due to differences in SSN enumeration methods. Third, the ACS has a relatively large sample: 3.5 million addresses in ACS versus 100,000 in the CPS ASEC and 14-65,000 in the SIPP. This means that relatively small groups, such as DI applicants, can be studied with the ACS. Fourth, the ACS surveys group quarters, which the CPS ASEC and SIPP do not. Fifth, the ACS has a higher response rate than other Census Bureau surveys: 95.8 percent in 2015 versus 76.3 percent for CPS ASEC and 75.7 percent for SIPP. This higher response rate makes nonresponse bias less worrisome (Meyer, Mok, and Sullivan 2015).

Data

This paper uses the 2015 ACS linked to three SSA administrative datasets: the 831 file, which tracks disability applications through Disability Determination Services (DDS) at the initial and reconsideration

[4] These are the OMB race and ethnicity standards implemented as of the date of this publication and as of the date of data collection of the 2015 ACS. More recently, OMB has updated those standards as described in Updates to Race/Ethnicity Standards for Our Nation

levels; the Detailed Earnings Record (DER), which captures earnings and self-employment income; and the Numident, which contains the applicant's date of birth.

Census Bureau researchers probabilistically link survey respondents to a Protected Identification Key (PIK) using identifying information such as name, date of birth, sex, and residential address (Wagner and Layne 2014). Once respondents are matched to a PIK, they can be linked to administrative records.

This paper examines disability allowance decisions at the initial and reconsideration levels for applications submitted in 2015-2019 by workers aged 25-65 at application date and applying under their own earnings record. Our analysis includes applicants in the US states and District of Columbia using ACS-reported state. We examine Title II disability applications, including those applying for concurrent SSI benefits, but not applications only for SSI, which we leave for future research. Our analysis does not examine applications that are denied in the nonmedical review stage, typically due to incomplete documentation, insufficient lifetime work history, or current employment. These "technical denials" do not reach disability determination review and are not included in our administrative data. Our analysis does not examine appeals to an Administrative Law Judge (ALJ) or higher levels since these appeals are also not captured by the 831 data. These applicants are included in our analysis but only up to the reconsideration level. We include individuals with previous applications or receipt spells before 2015 but only consider their applications started from 2015 to 2019. For individuals with multiple applications during the period examined, we consider only the first application. Appeals are not counted as separate applications.

We examine four race and ethnicity categories: NH Black, Hispanic of any race, NH Asian, and NH AIAN. We compare these groups to NH White applicants, separately by sex, using ACS-reported sex.

Our outcome variable is DI benefit allowance or denial, which we take from the 831 data. Our main explanatory variable is ACS-reported race and ethnicity.

We construct covariates from the ACS and administrative data. From the ACS, we construct controls for education (five categories), marital status (four categories), Census region (four categories), rural versus urban status, and a group quarters indicator; each of these variables is measured as of the ACS interview date, which occur throughout 2015. From the 831 administrative data, we take an indicator for a concurrent SSI claim and 12 dummy variables for body system codes. We also use fixed effects for application year spanning 2015-2019.

From the SSA DER file, we measure pre-disability earnings with the log of average, CPI-U-adjusted earnings from year (t-2) to year (t-6), where t=0 is the year of the application date. Earnings include wages, tips, and self-employment income. For those individuals with zero earnings in all five years, we code log earnings as zero and flag these values with a dummy variable.

Using the Numident date of birth and the 831 file's application date, we calculate the applicant's age at initial application. In our models, we include continuous age as well as dummy variables for ages 45-49, 50-54, and 55-65, since evaluators assess the applicant's education and the potential for retraining differently for these age groups.

Some records from the ACS cannot be linked to a PIK and therefore cannot be linked to the 831 and DER administrative data. The linkage rate for ACS respondents aged 25-65 is 94.4 percent. To account for selection into record linkage, we adjust the ACS replicate weights with inverse probability weighting.[5]

Methods

Prior research on differences in DI allowances by race and ethnicity has used descriptive statistics and logistic regression models (GAO 1992, 2004; Yang and Wang 2023). In this paper, we use these same methods but also implement propensity score matching (PSM) and inverse probability weighting (IPW). Whereas regression analyses estimate relationships between an outcome and explanatory variables across dissimilar cases, PSM and IPW compare outcomes across similar groups. PSM and IPW allow us to balance observed characteristics between groups so that we can contrast groups that differ by race and ethnicity but otherwise are similar in socioeconomic circumstances. PSM and IPW are most often used in analyses that make quasi-experimental inferences from observational data. However, we use these methods as a descriptive tool (Morgan and Winship 2014, Li and Li 2023). For brevity, we use causal terms such as "treatment" and "control" in our description of these methods, but our analysis is descriptive, not causal.

Matching involves identifying a "treatment group" (here those applicants whose race and ethnicity category is NH Black, Hispanic, NH Asian, or NH AIAN) and constructing a counterfactual "control" group (here NH White applicants) from the data that is similar on observable characteristics. After constructing this counterfactual group, one simply compares the outcome's group means, sometimes with an additional regression adjustment if substantial group differences remain after matching. Matching methods have the advantage that they focus on comparisons between overlapping groups, discarding treatment or control observations where a similar match cannot be found. Statistical results from other methods, including regression and inverse probability weighting, may be strongly influenced by treatment observations that are dissimilar to control observations, where there is little overlap.

PSM, the matching method used here, involves estimating the propensity of treatment assignment among all potential treatment and control observations and then matching observations with similar propensity scores. Rosenbaum and Rubin (1983) show that matching on propensity scores, if there is overlap, can achieve covariate balance between the treatment and control groups. PSM contrasts with blocking or exact matching, methods which match observations based on their values for covariates rather than the propensity score. The main advantage to PSM over these other matching methods is that PSM allows analysts to consider a greater number of covariates, such as the 12 body system codes in the 831 application file and the many observed characteristics in ACS. Attempting to match on all these variables themselves, rather than the propensity score, would result in few matches, if any.

Informed by recent debates on PSM (King and Nielsen 2019, Jann 2017b), we use a kernel matching algorithm for our main analysis. Kernel matching uses all overlapping control observations and weights them by their distance from the treatment observation's propensity score. We also performed nearest neighbor matching, but we found that it produced substantially greater covariate imbalance, even when we applied a caliper.

[5] Specifically, we use logistic regression to model PIK receipt as a function of ACS demographic, economic, and geographic characteristics. We then estimate the propensity score for each ACS respondent and multiply the corresponding ACS survey weight by the inverse of the estimated propensity score.

In addition to the matching methods, we use IPW, another propensity score method. Instead of matching on the propensity score, IPW reweights observations on the propensity score. IPW includes non-overlapping observations that are omitted in the PSM analyses.

Additionally, we perform logistic regression, which has been used in prior analyses of racial and ethnicity inequality in disability allowance rates (GAO 1992, 2004). Coefficients from logistic regression are conceptually dissimilar to the between-group differences estimated in PSM and IPW, so we use G-computation to estimate these between-group differences using regression models (Naimi, Cole, and Kennedy 2016; Wang, Nianogo, and Arah 2017; Greifer and Stuart 2021). Taking the comparison of Hispanic and NH White men as an example, G-computation involves estimating a regression model for all observations, subsetting just the observations for Hispanic men, predicting allowance for these observations where Hispanic ethnicity is true or false, and finally estimating between-group differences in allowance rates given the regression adjustment.

For all analyses, we use the ACS replicate weights adjusted for selection into administrative record linkage. For our PSM and IPW analyses, we use the kmatch package in Stata (Jann 2017a), which accommodates the ACS replicate weights.[6] For G-computation, we wrote a SAS program to do this analysis using the replicate weights.

Results

We begin our multivariate analysis by performing logistic regression with the linked ACS-administrative data. Table 3 reports odds ratios from logistic regressions of disability allowances on race and ethnicity and other covariates. These regressions compare NH Black vs. NH White, Hispanic vs. NH White, NH Asian vs. NH White, and NH AIAN vs. NH White, separately for each comparison and separately for male and female ACS respondents. Except for the continuous earnings and age variables, all variables are binomial. The odds ratios for these variables can be easily compared to assess the relative magnitude of their correlation with disability allowance. The odds ratios for race and ethnicity are statistically significant only for NH Black males relative to NH White males. In other groups, the race and ethnicity odds ratios are not statistically significant. Across all regressions, body system codes, age at application, and earnings have considerably larger and statistically significant associations with allowance than the race and ethnicity odds ratios. The largest odds ratios are for the genito-urinary system and neoplastic diseases body system codes, which have odds ratios 5-10 times larger than musculoskeletal system afflictions, the reference category.

We turn now to PSM and IPW. A first step for the PSM and IPW analyses is to run a logistic regression for each race and ethnicity category (relative to NH White) on other covariates. These regressions, presented in Table 4, estimate a propensity score, which is then used to match (PSM) or reweight (IPW) observations across groups. Table 4 shows that there is a statistically significant relationship between race and ethnicity and several body system codes, earnings, marital status, and other individual characteristics. Taken together, the results show that the groups differ in many observable ways.

Table 5 reports matching rates for PSM. The matching analysis selects NH Black, Asian, AIAN, or Hispanic applicants and matches them to NH White applicants with similar propensity scores. Often in matching analyses, there are some observations with propensities scores beyond the region of common

[6] Some replicate weights (but not the base weight) have negative values, always very few and of small magnitude. We recoded these to zero since neither kmatch nor any other matching package accommodates negative replicate weights. In regression analyses, we compared our results using negative weights and weights with negative values bottom coded to zero. The standard errors were identical to 20 significant digits.

support where propensity scores overlap. For example, suppose the top 10% of observations in the propensity score distribution are all the same race and ethnicity; these observations cannot be matched since there are no overlapping observations from individuals in other race and ethnicity groups. Table 5 shows, however, that there is a high level of common support. More than 90 percent of NH Black, NH Asian, and Hispanic applicants are matched to NH White applicants, while NH AIAN match percentages are somewhat lower at 88.1 percent for males and 87.5 percent for females. A large share of NH White applicants are used as matches, which shows that there is substantial overlap among all groups.

An important diagnostic step for propensity score matching and inverse probability weighting is assessing covariate balance in the matched or reweighted data. To assess differences across matched groups, two statistics are commonly used: the standardized mean difference (SMD) and variance ratio. Generally, SMDs exceeding 0.1 are considered to be evidence of covariate imbalance, though some consider higher thresholds of 0.2 (Zhang et al. 2019) to be acceptable. Variance ratios below 2 are generally considered acceptable.

While PSM and IPW analyses typically report covariate balance statistics for covariates, we do not do so in this paper. Since we examine two different propensity score methods with 35 covariates for eight groups, full covariate balance tables would report more than 3,000 additional cells, which would be prohibitive. Instead, we report in Table 6 the maximum standardized mean difference and variance ratio for each analysis.

Table 6 shows that in almost all PSM and IPW analyses, the maximum SMD and variance ratios are well below the thresholds that would suggest covariate imbalance. For all but one analysis, maximum SMDs are less than 0.1, and all variance ratios are less than 2. An exception is the PSM analysis for NH AIAN males, where the maximum SMD is 0.103. This corresponds to the dummy variable for endocrine body system code. Table 2 shows that applications due to endocrine system afflictions (diabetes for example) are more than twice as frequent among NH AIAN males relative to other males, and the SMD of 0.103 in Table 6 suggests that the PSM analysis cannot fully adjust for this difference. However, we note that applications due to endocrine system diseases are a relatively small share of applications for all groups. Furthermore, the next highest SMD for NH AIAN males is 0.054, which indicates that the PSM groups are otherwise well-balanced. Additionally, we estimated PSM with a regression adjustment and found results similar to our reported results.

Table 7 shows allowance rates by group and between-group differences. These allowance rates are estimated separately without adjustment, after PSM, after IPW, and after regression adjustment. The “no adjustment” row reports the raw group estimates and differences in allowance rates as observed in the matched ACS-administrative data. Before adjustment, there are notable differences in allowance rates between groups. The largest difference (-0.133) is for NH AIAN males. All other groups also have statistically significant differences, and in all instances, these allowance rates are lower than NH White applicants.

The other rows in Table 7 estimate the difference between observed NH Black, Hispanic, NH Asian, or NH AIAN applicants and a counterfactual group of NH White applicants whose observed characteristics are similar. With each adjustment method, the between-group differences are substantially attenuated, and in most instances, are not statistically significant. For seven of eight groups -- Hispanic males, NH Asian males, NH AIAN males, and females of all race and ethnicity groups -- none of the estimated differences are statistically significant after any adjustment method. For example, the largest observed raw difference is -0.133 for NH AIAN males relative to NH White males, which is not statistically significant after regression, PSM, or IPW.

For NH Black males relative to NH White counterfactual groups, differences remain statistically significant at the for regression and IPW. However, the differences estimated using PSM are not statistically significant.

Robustness checks

In addition to the four estimation methods presented so far, we performed other robustness checks. We estimated IPW models that used only observations included in the matching analysis, therefore excluding observations in the tails of the propensity score distribution, where propensity scores did not overlap. We found little difference in covariate balance or estimated allowance rates, which is expected given the high match rates. We also performed PSM using nearest-neighbor matching. The estimated allowance rates were similar, but covariate balance was somewhat worse, though generally acceptable.

We estimated PSM, IPW, and logistic regression models that included dummy variables for one-digit occupation. These data are self-reported free text fields in the ACS questionnaire which are then coded by the Census Bureau. We found that including occupational dummy variables did not meaningfully affect estimated differences or covariate balance. The direction, magnitude, and statistical significance patterns are similar with or without the inclusion of occupational dummies.

Conclusion

Using the 2015 ACS linked to SSA administrative records, our analysis examines differences in disability allowance rates by race and ethnicity, separately by sex. While there are sizable and statistically significant differences in raw allowance rates by race and ethnicity, these differences are substantially attenuated in multivariate analyses. For most groups, differences relative to NH White applicants of the same sex are not statistically significant using PSM, IPW, and regression G-computation. One exception is that we find statistically significant, negative differences for NH Black males relative to NH White males using IPW and regression G-computation, but this finding is not statistically significant when estimated with PSM.

That the differences by race and ethnicity are substantially attenuated after matching, reweighting, or regression adjustment demonstrates that raw differences in allowance rates are largely attributable to other observable differences—such as marital status, health conditions, and pre-disability earnings—rather than race and ethnicity. Yet for NH Black males, relative to matched NH White men, there is some evidence, albeit sensitive to estimation method, that racial differences remain after adjustment.

Our analysis complements earlier analyses (GAO 1992, 2004) by examining a more recent period, and we improve upon this prior work in two ways. First, prior analyses have used SSA administrative data on race, which do not conform to OMB standards for race and ethnicity data. Moreover, these data are shaped by selection into whether they are collected and updated. Instead, we use race and ethnicity data from the ACS, which are not affected by these issues. Second, by using the ACS's data on the social and economic circumstances of respondents, we incorporate richer contextual data into our analyses, such as the applicant's marital status and education, which may shape the propensity to apply, and in turn, the allowance rate.

There are three notable caveats of our analysis. One is that our analysis only examines initial decisions and reconsiderations. Higher appeals, such as to an administrative judge or appellate court, are not captured by the data that we can match to the ACS. Disability denials at the reconsideration level are frequently appealed at higher levels, and disability benefit receipt is often allowed by these judges and

courts. As a result, some share of the denials examined in this analysis are, or may be, appealed again and eventually allowed.

A second caveat is that our analysis is based on observational data—a survey linked to administrative data. To the extent that the variables we include in our analysis fail to account for the full heterogeneity between groups in the factors shaping DI application and allowance, then our estimates of the differences between groups may be impacted. Actual differences in the population may be larger or smaller than what we estimate. This is a limitation of all observational studies. With that said, our analysis considers many more potential correlates of DI allowance than prior studies using only administrative data.

A third caveat is that surveys may be affected by nonresponse bias. There has been growing attention to nonresponse bias in surveys, especially since the pandemic (Meyer, Mok, and Sullivan 2015; Rothbaum and Bee 2021; Rothbaum et al. 2021). The 2015 ACS response rate is quite high at 95.2 percent, but if disability applicants are concentrated among nonresponding households or persons, this may bias the results of our analysis. The challenge of nonresponse bias may be greater for future analyses, as household response rates have declined in more recent years.

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Table 1: Disability prevalence, application, and allowance rates by race and ethnicity, age 25-65, 2015-2019

	Non-Hispanic White	Non-Hispanic Black	Hispanic of any race	Non-Hispanic Asian	Non-Hispanic AIAN	All
Self-reported disability in 2015 ACS	11.9%	15.6%	9.7%	4.7%	20.2%	11.6%
DI application 2015-2019	3.0%	5.0%	2.7%	1.1%	4.2%	3.1%
DI allowance among applicants	42.8%	35.5%	36.9%	47.7%	33.3%	40.5%
DI allowance among population	1.3%	1.8%	1.0%	0.5%	1.4%	1.3%

Source: 2015 ACS matched to 2015-2019 SSA administrative records

Notes: Self-reported disability may not be work limiting. Reported estimates are approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008).

Table 2: Disability applicant characteristics, age 25-65, 2015-2019

	Male						Female					
	Non-Hispanic White	Non-Hispanic Black	Hispanic of any race	Non-Hispanic Asian	Non-Hispanic AIAN	All	Non-Hispanic White	Non-Hispanic Black	Hispanic of any race	Non-Hispanic Asian	Non-Hispanic AIAN	All
Mean earnings (t-2 to t-7)	$43,510	$31,450	$33,670	$42,640	$30,780	$39,820	$30,130	$25,670	$24,480	$46,570	$22,320	$28,590
Mean earnings (t-2 to t-7) (logged)	9.71	9.43	9.56	9.84	9.01	9.64	9.38	9.47	9.31	9.68	8.95	9.39
Zero mean earnings (t-2 to t-7)	0.05	0.05	0.05	0.04	0.09	0.05	0.05	0.03	0.04	0.06	0.06	0.05
Concurrent SSI claim	0.34	0.50	0.42	0.28	0.53	0.38	0.34	0.55	0.46	0.22	0.52	0.40
Mean age	50.8	48.5	48.8	52.0	49.0	50.1	49.6	47.2	48.1	52.1	49.5	48.9
Categorical age												
Age 25-44	0.25	0.33	0.32	0.24	0.32	0.28	0.29	0.38	0.34	0.21	0.28	0.32
Age 45-49	0.11	0.13	0.12	0.08	0.13	0.11	0.12	0.13	0.13	0.09	0.15	0.12
Age 50-54	0.17	0.18	0.17	0.15	0.18	0.17	0.18	0.16	0.18	0.20	0.17	0.18
Age 55-65	0.47	0.36	0.38	0.53	0.37	0.44	0.40	0.32	0.34	0.50	0.40	0.38
Education												
Less than high school	0.13	0.15	0.33	0.19	0.20	0.17	0.10	0.14	0.31	0.18	0.16	0.14
High school	0.42	0.42	0.35	0.23	0.35	0.40	0.35	0.33	0.28	0.20	0.28	0.33
Some college	0.32	0.33	0.24	0.30	0.39	0.31	0.39	0.40	0.31	0.26	0.47	0.38
Bachelor's degree	0.10	0.08	0.06	0.19	0.04	0.09	0.11	0.08	0.07	0.26	0.07	0.10
Advanced degree	0.04	0.03	0.02	0.09	0.02	0.04	0.05	0.03	0.02	0.10	0.02	0.04
Region												
Northeast	0.18	0.14	0.16	0.18	0.04	0.17	0.17	0.13	0.20	0.18	0.04	0.17
Midwest	0.25	0.17	0.09	0.13	0.15	0.21	0.26	0.19	0.09	0.13	0.20	0.22
South	0.40	0.61	0.37	0.23	0.40	0.43	0.40	0.60	0.36	0.21	0.38	0.43
West	0.17	0.09	0.38	0.46	0.42	0.19	0.16	0.07	0.35	0.49	0.38	0.18
Group quarters	0.02	0.04	0.03	0.02	0.04	0.03	0.01	0.01	0.01	0.01	0.01	0.01
Urban area	0.68	0.88	0.92	0.96	0.50	0.76	0.70	0.90	0.93	0.94	0.52	0.78
Marital status												
Married	0.51	0.38	0.51	0.68	0.44	0.49	0.48	0.26	0.42	0.61	0.38	0.42
Never married	0.23	0.37	0.29	0.20	0.32	0.26	0.18	0.44	0.27	0.17	0.29	0.26
Divorced	0.22	0.16	0.15	0.07	0.18	0.19	0.25	0.18	0.20	0.15	0.24	0.23
Separated	0.03	0.07	0.04	0.02	0.02	0.04	0.04	0.08	0.07	0.03	0.04	0.06
Widowed	0.02	0.02	0.01	0.02	0.03	0.02	0.04	0.04	0.04	0.05	0.05	0.04
Body system code												
1 Musculoskeletal system	0.39	0.37	0.40	0.27	0.39	0.38	0.37	0.38	0.41	0.33	0.39	0.38
2 Special senses and speech	0.02	0.03	0.03	0.05	0.01	0.03	0.02	0.02	0.03	0.04	0.02	0.02
3 Respiratory system	0.04	0.03	0.02	0.03	0.04	0.04	0.05	0.04	0.02	0.02	0.05	0.04
4 Cardiovascular system	0.09	0.10	0.09	0.11	0.08	0.09	0.05	0.07	0.04	0.04	0.04	0.05
5 Digestive system	0.03	0.02	0.03	0.02	0.03	0.03	0.03	0.02	0.02	0.02	0.03	0.02
6 Genito-urinary system	0.02	0.04	0.03	0.06	0.03	0.02	0.01	0.03	0.02	0.03	0.04	0.02
9 Endocrine system	0.02	0.03	0.04	0.02	0.06	0.03	0.02	0.03	0.03	0.02	0.04	0.02
11 Neurological	0.10	0.09	0.09	0.12	0.10	0.10	0.10	0.09	0.09	0.11	0.09	0.09
12 Mental disorders	0.15	0.15	0.15	0.15	0.13	0.15	0.18	0.16	0.16	0.13	0.12	0.17
13 Neoplastic diseases, malignant	0.10	0.06	0.07	0.10	0.07	0.09	0.10	0.07	0.07	0.17	0.05	0.09
14 Immune deficiency	0.01	0.03	0.02	0.01	0.01	0.02	0.03	0.03	0.03	0.03	0.04	0.03
20 Other or unknown code	0.05	0.05	0.05	0.05	0.06	0.05	0.06	0.05	0.07	0.05	0.09	0.06
Year												
2015	0.24	0.25	0.25	0.22	0.23	0.25	0.25	0.25	0.25	0.22	0.25	0.25
2016	0.22	0.21	0.20	0.19	0.19	0.21	0.22	0.22	0.22	0.21	0.21	0.22
2017	0.19	0.20	0.20	0.21	0.19	0.19	0.20	0.19	0.20	0.19	0.20	0.20
2018	0.18	0.18	0.17	0.20	0.21	0.18	0.17	0.17	0.17	0.20	0.19	0.17
2019	0.18	0.16	0.17	0.18	0.19	0.17	0.16	0.17	0.16	0.18	0.15	0.16
Unweighted N	29,000	5,400	4,300	800	600	40,000	27,500	7,200	4,400	800	600	40,500

Source: 2015 ACS matched to 2015-2019 SSA administrative records

Notes: Unweighted N rounded for disclosure avoidance. Reported estimates are approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008).

Table 3: Logistic regression of disability allowance on covariates, age 25-65, 2015-2019

	Male								Female							
	Non-Hispanic Black - Non-Hispanic White		Hispanic of any race - Non-Hispanic White		Non-Hispanic Asian - Non-Hispanic White		Non-Hispanic AIAN - Non-Hispanic White		Non-Hispanic Black - Non-Hispanic White		Hispanic of any race - Non-Hispanic White		Non-Hispanic Asian - Non-Hispanic White		Non-Hispanic AIAN - Non-Hispanic White	
Variable	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error
Non-Hispanic Black	0.879**	0.043							0.993	0.046						
Hispanic of any race			1.056	0.060							0.933	0.049				
Non-Hispanic Asian					0.984	0.131							0.834	0.088		
Non-Hispanic AIAN							0.834	0.128							0.978	0.126
Mean earnings (t-2 to t-7) (logged)	1.451***	0.025	1.409***	0.022	1.426***	0.024	1.420***	0.023	1.428***	0.027	1.431***	0.028	1.450***	0.028	1.452***	0.028
Zero earnings	2.508***	0.644	2.155**	0.527	2.054*	0.565	1.861*	0.504	3.026***	0.702	3.372***	0.888	3.758***	0.932	3.595***	0.893
Concurrent SSI claim	0.841***	0.036	0.916*	0.040	0.856***	0.038	0.877**	0.039	0.938	0.040	0.905*	0.039	0.927	0.046	0.918	0.047
Mean age	1.012*	0.006	1.015*	0.007	1.018**	0.007	1.017*	0.007	1.031***	0.005	1.023***	0.006	1.029***	0.006	1.031***	0.006
Categorical age																
Age 25-44 (omitted)																
Age 45-49	1.147	0.105	1.099	0.106	1.092	0.105	1.119	0.108	1.113	0.091	1.124	0.097	1.113	0.102	1.128	0.106
Age 50-54	2.179***	0.224	2.184***	0.250	1.964***	0.221	2.056***	0.229	1.797***	0.160	2.123***	0.213	1.869***	0.214	1.891***	0.208
Age 55-65	5.797***	0.786	5.709***	0.904	5.307***	0.817	5.579***	0.862	3.516***	0.401	4.261***	0.553	3.675***	0.530	3.639***	0.511
Education																
Less than high school	0.920	0.050	0.913	0.050	0.876*	0.050	0.883*	0.049	1.001	0.050	0.989	0.053	0.993	0.065	0.997	0.066
High school (omitted)																
Some college	0.803***	0.035	0.832***	0.033	0.811***	0.037	0.812***	0.037	0.850***	0.033	0.855***	0.035	0.855***	0.036	0.851***	0.035
Bachelor's degree	0.710***	0.043	0.755***	0.051	0.710***	0.047	0.720***	0.046	0.962	0.059	0.979	0.058	0.958	0.063	0.959	0.064
Advanced degree	0.602***	0.065	0.614***	0.066	0.582***	0.064	0.582***	0.069	0.845*	0.055	0.932	0.069	0.858*	0.061	0.907	0.067
Region																
Northeast (omitted)																
Midwest	0.988	0.044	0.992	0.048	0.981	0.049	0.988	0.050	0.804***	0.041	0.855**	0.047	0.797***	0.043	0.800***	0.045
South	0.825***	0.038	0.784***	0.037	0.779***	0.042	0.781***	0.042	0.671***	0.031	0.685***	0.037	0.629***	0.032	0.634***	0.034
West	0.742***	0.046	0.739***	0.042	0.701***	0.044	0.716***	0.048	0.713***	0.045	0.748***	0.052	0.683***	0.047	0.704***	0.050
Group quarters	0.750*	0.096	0.751	0.126	0.754	0.110	0.724*	0.107	0.925	0.286	1.261	0.372	1.378	0.436	1.247	0.409
Urban area	0.999	0.038	0.997	0.040	0.987	0.038	0.986	0.038	0.960	0.035	0.966	0.039	0.970	0.040	0.969	0.039
Marital status																
Married (omitted)																
Never married	1.061	0.049	1.012	0.049	1.106	0.061	1.079	0.059	1.213***	0.061	1.279***	0.064	1.271***	0.067	1.305***	0.069
Divorced	0.905*	0.039	0.829***	0.041	0.881*	0.047	0.875*	0.047	0.961	0.045	0.970	0.043	0.983	0.047	0.991	0.046
Separated	0.704***	0.065	0.758*	0.081	0.728**	0.084	0.700**	0.083	0.985	0.084	1.071	0.114	0.926	0.098	0.952	0.103
Widowed	1.084	0.163	1.089	0.168	1.170	0.187	1.226	0.201	0.998	0.072	1.028	0.084	0.994	0.080	0.974	0.076
Body system code																
1 Musculoskeletal system (omitted)																
2 Special senses and speech	2.071***	0.249	1.713***	0.204	1.755***	0.233	1.713***	0.237	3.087***	0.394	2.632***	0.391	2.483***	0.385	2.732***	0.419
3 Respiratory system	2.209***	0.165	2.364***	0.202	2.300***	0.187	2.245***	0.186	3.065***	0.199	3.009***	0.275	3.358***	0.302	3.269***	0.278
4 Cardiovascular system	1.468***	0.106	1.415***	0.112	1.350***	0.101	1.345***	0.105	1.462***	0.115	1.271*	0.129	1.350**	0.128	1.360**	0.130
5 Digestive system	1.757***	0.220	1.937***	0.248	1.853***	0.237	1.899***	0.250	1.478***	0.165	1.553***	0.184	1.549***	0.194	1.593***	0.195
6 Genito-urinary system	11.38***	1.850	10.93***	1.973	8.993***	1.689	8.661***	1.767	7.456***	1.086	4.761***	0.830	4.637***	0.807	4.695***	0.842
9 Endocrine system	0.504***	0.062	0.552***	0.075	0.525***	0.074	0.567***	0.079	0.399***	0.055	0.402***	0.058	0.360***	0.057	0.389***	0.063
11 Neurological	2.357***	0.117	2.450***	0.129	2.239***	0.128	2.254***	0.129	2.550***	0.161	2.464***	0.173	2.578***	0.184	2.602***	0.193
12 Mental disorders	1.750***	0.101	1.678***	0.101	1.635***	0.109	1.646***	0.108	1.614***	0.077	1.510***	0.076	1.589***	0.079	1.579***	0.079
13 Neoplastic diseases, malignant	9.964***	0.906	9.814***	0.957	10.45***	1.003	10.45***	1.016	9.312***	0.618	9.298***	0.722	9.389***	0.675	10.04***	0.737
14 Immune deficiency	1.584***	0.184	1.548***	0.186	1.494**	0.200	1.495**	0.206	1.673***	0.161	1.487***	0.148	1.598***	0.171	1.645***	0.174
20 Other or unknown code	0.567***	0.066	0.582***	0.061	0.557***	0.074	0.598***	0.078	0.691***	0.066	0.573***	0.063	0.603***	0.070	0.626***	0.074
Year																
2015 (omitted)																
2016	0.948	0.049	0.911	0.045	0.920	0.046	0.922	0.046	1.013	0.054	1.045	0.054	1.022	0.059	1.018	0.059
2017	0.876*	0.048	0.891*	0.047	0.877*	0.050	0.885*	0.050	1.048	0.052	1.061	0.053	1.040	0.059	1.043	0.058
2018	0.936	0.046	0.895	0.052	0.916	0.052	0.926	0.052	1.053	0.056	1.115	0.063	1.072	0.066	1.087	0.067
2019	0.898*	0.047	0.850**	0.046	0.894*	0.050	0.878*	0.051	1.196**	0.069	1.213**	0.080	1.198**	0.076	1.213**	0.077
Unweighted N	34,000		33,000		29,500		29,000		34,500		32,000		28,000		28,000	

Source: 2015 ACS matched to 2015-2019 SSA administrative records

Notes: *** p<.001, ** p<.01, * p<.05. Standard errors are calculated using ACS replicate weights adjusted for differential probability of administrative record linkage. Reported estimates are approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008).

Table 4: Logistic regression of race and ethnicity on covariates, age 25-65, 2015-2019

		Male								Female							
		Non-Hispanic Black - Non-Hispanic White		Hispanic of any race - Non-Hispanic White		Non-Hispanic Asian - Non-Hispanic White		Non-Hispanic AIAN - Non-Hispanic White		Non-Hispanic Black - Non-Hispanic White		Hispanic of any race - Non-Hispanic White		Non-Hispanic Asian - Non-Hispanic White		Non-Hispanic AIAN - Non-Hispanic White	
Variable		Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error	Odds ratio	Standard error
Mean earnings (t-2 to t-7) (logged)		0.929***	0.018	0.988	0.021	0.906**	0.033	0.915	0.044	1.086***	0.018	1.034	0.023	1.197***	0.059	0.910*	0.039
Zero earnings		0.392***	0.082	0.746	0.176	0.284**	0.116	0.524	0.265	1.419*	0.240	0.990	0.238	7.256***	3.897	0.401	0.210
Concurrent SSI claim		1.490***	0.064	1.144**	0.058	0.999	0.137	1.761***	0.202	1.714***	0.078	1.340***	0.069	0.825	0.104	1.834***	0.256
Mean age		1.003	0.007	0.988*	0.006	1.014	0.015	1.020	0.018	1.010	0.006	0.998	0.007	1.016	0.014	1.017	0.018
Categorical age																	
	Age 25-44 (omitted)																
	Age 45-49	0.984	0.118	0.982	0.109	0.724	0.172	0.796	0.212	0.978	0.078	1.031	0.110	0.826	0.190	1.263	0.361
	Age 50-54	1.021	0.136	0.972	0.107	0.776	0.226	0.713	0.227	0.842	0.090	0.909	0.123	1.092	0.318	0.907	0.304
	Age 55-65	0.849	0.141	0.898	0.132	0.781	0.290	0.509	0.215	0.791	0.106	0.810	0.140	0.912	0.321	1.060	0.447
Education																	
	Less than high school	1.012	0.066	3.125***	0.161	2.941***	0.395	1.515*	0.249	1.239***	0.070	3.624***	0.211	3.740***	0.567	1.804***	0.311
	High school (omitted)																
	Some college	1.037	0.051	0.765***	0.040	1.302	0.177	1.403*	0.189	1.066	0.045	0.869**	0.041	0.943	0.136	1.519***	0.185
	Bachelor's degree	0.798**	0.065	0.598***	0.053	2.453***	0.377	0.586	0.176	0.805**	0.056	0.683***	0.057	2.764***	0.384	0.916	0.223
	Advanced degree	0.750*	0.101	0.547***	0.069	2.533***	0.444	1.008	0.366	0.754**	0.071	0.429***	0.055	1.969**	0.425	0.634	0.236
Region																	
	Northeast (omitted)																
	Midwest	0.876	0.059	0.381***	0.033	0.568***	0.084	2.234	0.961	0.985	0.048	0.311***	0.023	0.536***	0.083	2.947***	0.859
	South	2.317***	0.152	1.107	0.076	0.675**	0.095	3.344**	1.375	2.452***	0.120	0.857**	0.047	0.624**	0.088	3.431***	0.966
	West	0.616***	0.052	2.234***	0.164	2.516***	0.260	11.56***	4.778	0.564***	0.039	1.691***	0.096	2.781***	0.318	10.84***	3.177
Group quarters		1.406**	0.158	1.058	0.147	1.129	0.506	1.422	0.457	0.872	0.204	0.687	0.350	1.090	0.731	1.495	0.940
Urban area		4.118***	0.218	5.078***	0.372	9.033***	1.529	0.358***	0.045	4.547***	0.238	5.502***	0.419	5.170***	0.865	0.367***	0.045
Marital status																	
	Married (omitted)																
	Never married	1.646***	0.079	0.915	0.060	0.529***	0.074	1.308	0.219	3.461***	0.145	1.238**	0.088	0.734*	0.095	2.173***	0.280
	Divorced	0.794***	0.040	0.636***	0.046	0.236***	0.053	0.814	0.130	1.028	0.046	0.784***	0.052	0.439***	0.067	0.993	0.159
	Separated	2.043***	0.217	0.943	0.127	0.502*	0.156	0.580	0.185	2.409***	0.201	1.326**	0.128	0.621	0.197	1.147	0.321
	Widowed	1.208	0.169	0.748	0.212	0.963	0.276	2.039	0.915	1.540***	0.135	0.824	0.096	0.872	0.169	1.409	0.344
Body system code																	
1	Musculoskeletal system (omitted)																
2	Special senses and speech	1.336*	0.167	1.109	0.157	2.603***	0.537	0.726	0.275	1.110	0.127	1.293	0.167	2.122**	0.507	1.084	0.374
3	Respiratory system	0.645***	0.066	0.495***	0.090	1.060	0.458	0.938	0.306	0.746**	0.067	0.409***	0.067	0.546	0.187	0.966	0.260
4	Cardiovascular system	1.137	0.080	1.095	0.089	1.980***	0.337	0.979	0.219	1.360***	0.118	0.805	0.101	1.105	0.235	0.762	0.243
5	Digestive system	0.597**	0.108	0.937	0.156	1.088	0.379	1.193	0.405	0.474***	0.066	0.640**	0.100	0.876	0.306	1.024	0.298
6	Genito-urinary system	2.591***	0.317	1.515**	0.238	4.674***	1.000	1.712	0.537	2.179***	0.316	1.822**	0.346	3.428***	0.952	3.168***	0.916
9	Endocrine system	1.631***	0.225	1.861***	0.258	1.770*	0.502	3.114***	0.737	1.653***	0.194	1.791***	0.244	1.614	0.719	2.366**	0.617
11	Neurological	0.890	0.065	0.864	0.064	1.507**	0.215	1.145	0.237	0.928	0.061	0.832*	0.071	1.120	0.185	0.943	0.227
12	Mental disorders	0.788***	0.048	0.857	0.068	1.376	0.224	0.725	0.149	0.682***	0.038	0.720***	0.050	0.903	0.127	0.663	0.142
13	Neoplastic diseases, malignant	0.720***	0.056	0.786*	0.076	1.306	0.182	0.909	0.226	0.738***	0.054	0.733***	0.066	1.486**	0.195	0.673	0.194
14	Immune deficiency	1.590***	0.212	1.299	0.280	1.140	0.588	0.722	0.306	1.182	0.127	1.418*	0.190	1.151	0.318	1.772*	0.503
20	Other or unknown code	0.976	0.092	0.888	0.098	1.632*	0.341	0.938	0.253	0.793*	0.071	0.884	0.102	0.848	0.192	1.378	0.375
Year																	
	2015 (omitted)																
	2016	0.977	0.060	0.940	0.065	1.007	0.124	1.000	0.173	1.032	0.053	1.037	0.072	0.960	0.116	0.952	0.157
	2017	1.040	0.060	1.103	0.082	1.241	0.192	1.080	0.188	0.938	0.053	0.965	0.068	0.934	0.114	0.950	0.176
	2018	1.028	0.062	0.987	0.074	1.279	0.198	1.282	0.243	0.978	0.056	0.957	0.064	1.083	0.121	1.046	0.194
	2019	0.896	0.060	0.994	0.067	1.120	0.187	1.257	0.232	0.961	0.057	0.976	0.071	1.113	0.152	0.850	0.164
Unweighted N		34,000		33,000		29,500		29,000		34,500		32,000		28,000		28,000	

Source: 2015 ACS matched to 2015-2019 SSA administrative records

Notes: *** p<.001, ** p<.01, * p<.05. Standard errors are calculated using ACS replicate weights adjusted for differential probability of administrative record linkage. Reported estimates are approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008).

Table 5 -- Unweighted match percentage for propensity score matching

Sex	Group	Group match percentage	NH White match percentage	
Male	Non-Hispanic black	94.3%	80.0%	
	Hispanic of any race	93.4%	96.5%	
	Non-Hispanic Asian	91.2%	73.9%	
	Non-Hispanic AIAN	88.1%	84.5%	
Female	Non-Hispanic black	94.7%	85.1%	
	Hispanic of any race	94.1%	96.7%	
	Non-Hispanic Asian	93.6%	95.3%	
	Non-Hispanic AIAN	87.5%	91.9%	

Source: 2015 ACS matched to 2015-2019 SSA administrative records

Notes: Reported estimates are approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008).

Table 6: Maximum values for covariate balance diagnostics

		Propensity score matching		Inverse probability weighting	
Sex	Treatment group	Maximum standardized mean difference	Maximum variance ratio	Maximum standardized mean difference	Maximum variance ratio
Male	Non-Hispanic Black	0.052	1.21	0.030	1.11
	Hispanic of any race	0.051	1.26	0.055	1.09
	Non-Hispanic Asian	0.051	1.20	0.012	1.10
	Non-Hispanic AIAN	0.103	1.49	0.012	1.08
Female	Non-Hispanic Black	0.037	1.26	0.025	1.12
	Hispanic of any race	0.037	1.22	0.048	1.07
	Non-Hispanic Asian	0.034	1.13	0.015	1.06
	Non-Hispanic AIAN	0.067	1.35	0.011	1.06

Source: 2015 ACS matched to 2015-2019 SSA administrative records

Notes: Reported estimates are approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008).

Table 7 -- Mean differences before and after logistic regression, matching, or inverse probability weighting

	Male															
	Non-Hispanic Black - Non-Hispanic White comparison				Hispanic of any race - Non-Hispanic White comparison				Non-Hispanic Asian - Non-Hispanic White comparison				Non-Hispanic AIAN - Non-Hispanic White comparison			
Adjustment method	Non-Hispanic Black allowance rate	Non-Hispanic White allowance rate	Difference	Standard error	Hispanic allowance rate	Non-Hispanic White allowance rate	Difference	Standard error	Non-Hispanic Asian allowance rate	Non-Hispanic White allowance rate	Difference	Standard error	Non-Hispanic AIAN allowance rate	Non-Hispanic White allowance rate	Difference	Standard error
No adjustment	0.382	0.477	-0.095***	0.010	0.423	0.477	-0.054***	0.010	0.508	0.477	0.031***	0.022	0.343	0.477	-0.133***	0.024
Regression g-computation	0.375	0.396	-0.021**	0.008	0.424	0.415	0.009	0.010	0.511	0.514	-0.003	0.023	0.354	0.383	-0.029	0.024
Propensity score matching	0.383	0.410	-0.026	0.014	0.425	0.430	-0.005	0.027	0.505	0.519	-0.014	0.037	0.353	0.391	-0.037	0.045
Inverse probability weighting	0.382	0.399	-0.017*	0.008	0.423	0.404	0.019	0.010	0.508	0.507	0.001	0.023	0.343	0.381	-0.037	0.024

	Female															
	Non-Hispanic Black - Non-Hispanic White				Hispanic of any race - Non-Hispanic White				Non-Hispanic Asian - Non-Hispanic White				Non-Hispanic AIAN - Non-Hispanic White			
Adjustment method	Non-Hispanic Black allowance rate	Non-Hispanic White allowance rate	Difference	Standard error	Hispanic allowance rate	Non-Hispanic White allowance rate	Difference	Standard error	Non-Hispanic Asian allowance rate	Non-Hispanic White allowance rate	Difference	Standard error	Non-Hispanic AIAN allowance rate	Non-Hispanic White allowance rate	Difference	Standard error
No adjustment	0.324	0.370	-0.046***	0.008	0.309	0.370	-0.060***	0.01	0.435	0.370	0.065**	0.021	0.311	0.370	-0.059*	0.027
Regression g-computation	0.319	0.320	-0.001	0.007	0.306	0.318	-0.011	0.008	0.426	0.458	-0.031	0.018	0.310	0.314	-0.004	0.020
Propensity score matching	0.326	0.336	-0.010	0.013	0.318	0.327	-0.010	0.015	0.431	0.459	-0.029	0.025	0.298	0.319	-0.021	0.032
Inverse probability weighting	0.324	0.324	0.000	0.009	0.309	0.321	-0.012	0.010	0.435	0.464	-0.029	0.019	0.311	0.318	-0.007	0.021

Source: 2015 ACS matched to SSA administrative records

Notes: *** p<.001, ** p<.01, * p<.05. Standard errors are calculated using ACS replicate weights adjusted for differential probability of administrative record linkage. Reported estimates are approved for release by the Census Bureau's Disclosure Review Board (CBDRB-FY25-CES014-008).